

SEBES ALBA, Romania

WMO: 152310

Lat: 45.9643N

Lon: 23.5334E

Elev: 893

StdP: 14.23

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	13.7	4.8	7.3	9.3	10.1	9.5	16.4	27.4	20.6	23.5	29.0	2.7	130	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.6	90.6	68.5	87.3	68.3	84.2	67.4	71.8	84.2	70.4	82.6	69.1	80.7	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.1	106.5	76.3	66.5	100.8	74.8	65.3	96.6	73.5	36.1	84.3	34.8	81.5	33.7	81.3	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	19.2	16.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	28.9	35.5	42.9	51.8	61.3	67.7	69.6	70.4	62.3	51.6	41.9	33.3		
	DBStd	15.86	8.57	8.27	7.53	8.25	5.55	5.32	4.83	5.10	7.58	6.67	7.65	7.15		
	HDD50	2223	655	407	238	79	2	0	0	0	2	62	259	519		
	HDD65	5463	1120	827	686	398	139	33	17	14	139	415	692	983		
	CDD50	2773	0	1	17	131	353	530	607	634	371	112	17	0		
	CDD65	538	0	0	0	1	24	113	159	182	59	0	0	0		
	CDH74	5569	0	0	3	88	339	1127	1508	1874	608	22	0	0		
	CDH80	1856	0	0	0	5	49	339	504	764	195	0	0	0		
(14) Wind	WSAvg	5.4	5.6	5.9	6.3	6.5	5.6	4.8	5.1	4.2	5.2	5.0	4.8	6.3		
(15) Precipitation	PrecAvg	24.8	1.0	0.9	1.3	2.3	2.9	3.7	3.6	2.4	2.3	1.8	1.3	1.4		
	PrecMax	33.0	1.9	2.4	2.8	4.4	6.1	6.9	7.1	5.2	5.1	3.7	2.7	3.1		
	PrecMin	15.6	0.3	0.2	0.3	0.5	1.1	1.0	0.7	0.4	0.2	0.1	0.0	0.2		
	PrecStd	4.6	0.5	0.5	0.6	1.0	1.2	1.4	1.6	1.1	1.3	1.0	0.7	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	59.2	71.0	80.6	84.3	90.9	92.6	95.8	91.4	77.7	65.2	52.0		
		MCWB	44.2	49.5	52.8	60.3	64.3	71.0	69.1	69.4	68.9	61.5	55.9	46.6		
	2%	DB	44.9	54.3	64.6	76.5	81.1	87.1	89.5	91.3	85.6	73.0	59.6	48.0		
		MCWB	42.5	47.2	50.8	58.6	63.5	70.5	68.8	67.5	66.0	59.3	53.0	45.1		
	5%	DB	41.3	50.2	59.9	71.9	78.1	83.9	85.5	87.6	81.0	69.1	56.4	45.1		
		MCWB	39.6	44.9	48.4	56.8	62.7	68.4	68.1	67.2	63.6	57.9	51.5	43.0		
	10%	DB	38.9	47.0	55.2	67.7	74.4	80.9	81.9	83.9	76.8	64.3	53.1	42.2		
		MCWB	37.5	43.0	45.7	55.1	61.1	66.9	66.9	67.4	63.0	56.3	49.2	40.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	51.3	54.8	62.5	69.3	75.0	73.2	72.4	69.5	64.6	56.4	48.5		
		MCDB	47.8	58.4	69.0	75.2	79.3	85.3	84.5	86.5	88.1	73.1	64.5	50.0		
	2%	WB	42.9	48.2	51.5	60.2	66.2	72.3	71.4	70.8	67.3	61.0	54.2	45.3		
		MCDB	45.2	52.9	62.1	73.6	75.5	84.1	82.9	84.8	82.2	68.7	58.9	47.4		
	5%	WB	40.0	45.9	49.1	58.1	64.2	70.3	70.0	69.5	65.3	58.9	52.3	42.9		
		MCDB	41.3	49.8	58.1	70.6	73.7	81.1	81.5	82.7	77.0	67.1	56.4	44.9		
	10%	WB	37.7	43.3	47.0	55.9	62.3	68.5	68.6	68.4	63.6	56.7	49.7	40.8		
		MCDB	38.9	46.6	53.6	65.8	72.1	78.1	79.4	80.4	74.0	63.8	53.0	42.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	15.0	18.7	22.1	20.8	21.7	22.3	23.6	21.1	20.3	13.9	9.0		
		MCDBR	12.1	21.1	27.9	31.1	28.7	27.4	29.6	31.1	29.2	30.7	19.1	13.3		
	5% WB	MCWBR	9.8	14.1	15.7	15.2	13.2	11.6	11.2	10.7	10.6	16.9	12.3	10.4		
		MCWBR	12.2	18.9	24.3	28.2	23.5	23.9	25.0	26.1	25.8	26.9	17.6	13.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.323	0.354	0.401	0.463	0.447	0.440	0.460	0.471	0.423	0.381	0.351	0.323		
		taud	2.324	2.246	2.169	2.026	2.139	2.204	2.155	2.115	2.212	2.316	2.346	2.345		
	Ebn at Noon	Ebn at Noon	242	255	259	252	261	264	256	247	249	243	229	228		
		Edh at Noon	25	32	41	51	47	44	46	46	38	30	24	22		
(44) All-Sky Solar Radiation	RadAvg	403	655	1011	1373	1643	1832	1801	1651	1176	818	477	325			
	RadStd	52	83	143	148	154	167	156	161	127	96	62	42			

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)		+1.24	N/A	N/A	+1.71	N/A	N/A	-229	-342	+182	+81

Nomenclature: See separate page